VIETNAMESE - GERMAN UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE



# Customer Segmentation Analysis REPORT

## Customer and Segmentation Data Analytics

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| Nguyễn Hà Khải | Dimensions implementation, data warehouse design | 100% |
| Nguyễn Quang Ngọc Trác | K-means clustering, RFM technique exploration implementaion | 100% |
| Tô Huỳnh Phúc | Fact tables, schedulers, and incremental loading implementation | 100% |
| Nguyễn Hữu Triển | Topic, data source, and case studies investigation | 100% |

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# Chapter 1

# Introduction

## 1.1 Project Context and Data Source

Customer purchasing behaviour is an important input to retail decisions, yet transaction tables alone do not readily show which customers buy repeatedly, which customers generate high value, or which customer groups require different marketing approaches. CompanyX is the retail dataset used in this project. Its transactional schema records customers, sales orders, order details, products, and associated customer demographics. The available sales history begins in May 2011 and ends in June 2014.

The project deliberately focuses on *customer-centric analysis*, rather than general sales reporting. In particular, the team uses Recency, Frequency, and Monetary (RFM) behaviour to describe customers, K-Means to discover five interpretable groups, and a dimensional data warehouse to make these results available for analysis. The warehouse also stores slowly changing customer attributes and monthly historical observations. Its automated ETL pipeline makes the latest customer profiles and purchase-triggered segmentation results available at one-minute intervals.

## 1.2 Business Problem

The practical question is not simply how much CompanyX sold, but *which customers contributed that value and how their purchasing patterns changed.* Four issues motivated the solution:

- **Repeat purchasing and retention.** Single-order buyers and customers who do not return represent different opportunities from loyal repeat customers. An appropriate intervention requires a view of their purchase frequency and time since their last order.
- **Uneven customer value.** Customers differ considerably in their spending. Treating

all customers identically can hide high-value accounts and lead to inefficient campaign allocation.

- **Cross-selling opportunities.** Understanding a customer's value and engagement stage can help marketers decide when to recommend accessories, maintenance products, or complementary items. RFM identifies target groups; verifying product-specific cross-selling gaps would require separate product-level analysis.
- **Fragmented historical and operational views.** A record updated when a purchase occurs answers a different question from a month-end view of the entire customer population. Both are needed to distinguish purchase-time behaviour from longer-term inactivity and segment migration.

The Power BI dashboard provides key indicators such as repeat-customer rate, customer inactivity, average purchase frequency, and spending patterns. Together, these metrics offer an overview of customer purchasing behavior and engagement, helping decision-makers identify potential retention issues, understand customer value, and develop appropriate marketing strategies.

## 1.3 Objectives and Deliverables

The project focuses on two main goals:

1. **Customer Segmentation:** Analyse customer purchasing patterns to identify distinct customer groups based on their behaviour and value. These insights can help the company better understand its customer base and develop more effective marketing, retention, and customer engagement strategies.
2. **Automated Business Intelligence Pipeline:** Develop an automated system that help businesses focus on analysis with less time spent on technical configurations and setup.

To achieve these objectives, the project delivers an integrated solution consisting of a data warehouse, automated data processing pipelines, machine learning-based customer segmentation, and interactive business intelligence dashboards. Regarding these components, their results will be discussed thoroughly in following chapters.

# Chapter 2

# Background and Related Applications

## 2.1 Customer Segmentation with RFM and K-Means

RFM analysis summarizes purchasing behaviour using three variables: **Recency** (elapsed days since the most recent purchase), **Frequency** (number of orders), and **Monetary** value (amount spent). An RFM profile makes customer behaviour measurable without relying on arbitrary demographic categories. In this project, frequency and monetary values are calculated using a trailing 365-day window, while recency is measured relative to the observation date. The definitions and preprocessing are described in Chapter 5.

Traditional RFM segmentation commonly places customers into groups according to hand-written thresholds or ranks. This can be transparent and inexpensive, but the boundaries must be selected in advance. K-Means provides an alternative: customers are grouped according to their positions in a numeric feature space, using distances to cluster centroids. For this project, the feature space is the log-transformed and standardized RFM vector. K-Means does not automatically supply business labels; the clusters must be interpreted using their purchasing statistics.

The segmentation model alone is not sufficient for operational business intelligence. A *data warehouse* provides consistent customer keys, historical demographic versions, RFM snapshots, and persisted model configurations. SQL Server Integration Services (SSIS) performs incremental extraction and loading, while Power BI connects the results to decision-making dashboards. This combination is the central design choice of CompanyX.

## 2.2 Related Industry Examples

### 2.2.1 Black Diamond: Data-Driven Retention

The outdoor-equipment brand Black Diamond worked with customer-data platform provider Lexer to consolidate customer information and improve targeted acquisition and retention activities. According to the provider's case study, separate initiatives achieved a 50% reduction in acquisition cost and a 1,101% increase in revenue per email for a lapsed-customer campaign (Lexer 2019). These are externally reported business outcomes, *not* results of the CompanyX implementation. They illustrate why identifying inactive and valuable customers may support differentiated actions.

### 2.2.2 K-Means as a Customer Segmentation Method

A published customer-segmentation tutorial presents K-Means as a way to form groups from numerical customer features (Towards AI 2022). Its general lesson is relevant here: numerical scaling, cluster-number selection, and cluster interpretation are all necessary before the grouping is useful for marketing. A common demonstration uses annual income and a spending score, whereas *CompanyX uses RFM transaction features.* We do not treat the tutorial as evidence of an internal deployment by a particular retailer.

### 2.2.3 Alibaba Cloud: An End-to-End Segmentation Example

Pratama's Alibaba Cloud tutorial demonstrates a pipeline for customer segmentation using data preparation, MaxCompute storage, K-Means training, and model serving (Pratama 2020). The tutorial uses a public retail dataset as an example; it is *not evidence* that the author's employer, DANA Indonesia, operated the same pipeline in production. For CompanyX, the relevant principle is linking the data pipeline with a repeatable training and scoring workflow, rather than running clustering once in an isolated notebook.

## 2.3 Approach Comparison

## 2.4 Problem Statement and Chosen Solution

CompanyX needs an integrated process for answering three business questions: *Which customers are most valuable? Which customers show reduced engagement? How can the company tailor its actions to distinct purchasing behaviours?* The team addresses them with five RFM-based clusters, two complementary fact tables, and an automated path from operational records to a

Table 2.1: Comparison of approaches relevant to the project.

| **Approach** | **Typical implementation** | **Relevance and limitation** |
|---|---|---|
| Rule-based RFM | SQL thresholds or ranked RFM scores | Transparent and simple, but business thresholds are selected manually. |
| K-Means clustering | Unsupervised clustering of scaled numeric attributes | Learns groups from observed customer behaviour, but does not attach meaningful segment names or automate refresh by itself. |
| Warehouse + RFM + K-Means + BI | Incremental ETL, history, model configurations and dashboards | Combines purchase-time segmentation with retrospective customer analysis; requires coordinated loading, training, and reporting. |

Power BI dashboard. The resulting segments provide an analytical basis for recommendations; proving that a proposed marketing campaign increases revenue or reduces churn would require a separate intervention study.

# Chapter 3

# Data Warehouse Design

The proposed warehouse follows a **fact constellation (galaxy) schema**: two customer-centred fact tables share dimensions for customer profiles, historical demographic versions, model configuration, and calendar dates. The diagram in Figure 3.1 is the team's final warehouse design. A separate general-purpose sales fact is intentionally excluded because the project's analytical grain is the *customer*, not the sales line. The RFM procedures read the required transactions from the source `CompanyX` database.

## 3.1 Staging Layer

The customer staging table uses a composite primary key combining the batch ID and customer key. It receives the demographic records selected by SSIS before they are merged into the two profile dimensions. A new batch identifier distinguishes the records associated with a particular ETL run. The active package clears the staging table at the beginning of a run and passes the batch identifier to `usp_ApplyCustomerProfileBatch`.

This is a *relational staging area*, not a durable message queue or an archive of every ETL batch. Its batch identifier makes processing scoped and repeatable; transactional error handling in the merge procedure protects the consistency of the dimension changes. Keeping a long-term execution log or recovering an interrupted run from retained staging records would require additional mechanisms.

## 3.2 Dimension Layer and Slowly Changing Dimensions

### 3.2.1 DimCustomerProfiling

This dimension stores one current demographic row per customer. The source `CustomerID` field in `Sales.Customer` is converted to an `NVARCHAR(20)` value, `CustomerKey_Str`, which is

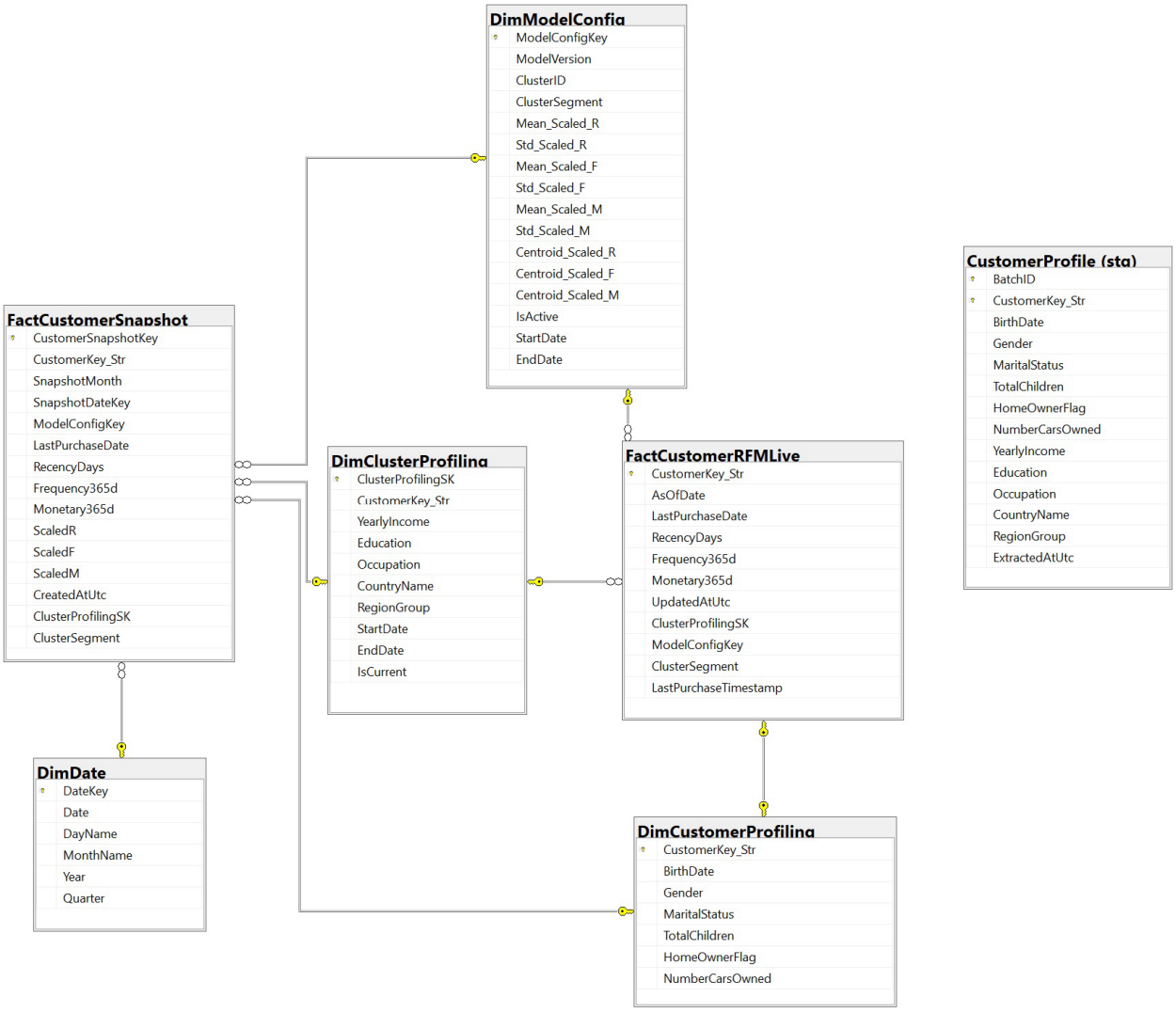



Figure 3.1: Final CompanyX customer-oriented fact constellation.

reused as the primary key and in the related fact tables. Birth date is handled as an **SCD Type 0** attribute: once populated, the original value is retained. Gender, marital status, number of children, home ownership, and number of cars are handled as **SCD Type 1** attributes: detected changes overwrite the current value instead of preserving previous versions. This design retains straightforward access to current customer information without unnecessarily multiplying records.

### 3.2.2 DimClusterProfiling

`DimClusterProfiling` stores socioeconomic and geographical characteristics including yearly income, education, occupation, country, and region. These are managed as **SCD Type 2** attributes. A changed profile closes the preceding version and inserts a new row with a distinct `ClusterProfilingSK`, effective dates, and an `IsCurrent` flag. This allows analysts to inspect demographic versions captured by the warehouse and associate customer RFM observations with the available historical profiles.

An important distinction is that this dimension's name refers to *customer profiling*, not the five K-Means segment definitions. It also cannot recover demographic history from before the first successful warehouse load; retrospective sales snapshots should not be interpreted as proof of what those demographic fields were in earlier years.

### 3.2.3 DimModelConfig

`DimModelConfig` persists the output of K-Means training. Each model version contains five rows, one for each cluster, with its numeric cluster ID, assigned business segment name, per-feature scaling means and standard deviations, and centroid coordinates in standardized feature space. `StartDate`, `EndDate`, and `IsActive` distinguish model versions over time. This resembles Type 2 effective-dated version management, but it represents *retrained models*, not changing customer demographics. Persisting the parameters lets the Python live scorer reuse a trained model without refitting K-Means for every customer.

### 3.2.4 DimDate

Each row of `DimDate` corresponds to a calendar date and exposes fields such as year, quarter, day, and month. The schema initialization script fills missing dates from **1 January 2010** through the date of execution plus twenty calendar years. The range covers the source sales period and the intended future reporting horizon; rerunning the script may extend it as needed.

## 3.3 Fact Layer

### 3.3.1 FactCustomerSnapshot

`FactCustomerSnapshot` is a **monthly periodic snapshot fact**. Its analytical grain is *one eligible customer at one month-end date*, uniquely identified by customer and `SnapshotDateKey`. The snapshot procedure reconstructs RFM from sales transactions as of that date. `Frequency365d` counts distinct purchases and `Monetary365d` sums sales order-line values within the trailing 365 days. `RecencyDays` measures the number of days since the latest purchase and can exceed 365 days. The fact also stores model references, a segment label, scaled features, and the associated customer profile key where available.

Snapshots are calculated independently from the transaction-triggered live table; therefore, their row counts do *not* necessarily match the live table. The monthly process is scheduled at 00:05 on the first day of the following month and uses the preceding calendar month-end as the snapshot date. This separates the historical *as-of-date* view from purchase-event scoring and makes it possible to observe inactivity even when a customer places no new order.

### 3.3.2 FactCustomerRFMLive

`FactCustomerRFMLive` maintains one record per customer rather than one record per order. When relevant sales changes are detected, the system recalculates the affected customer's orders and inserts or updates that record. The analysis date is intentionally set to the *customer's latest purchase date*; consequently, the purchase-time `RecencyDays` is zero. Frequency and monetary value summarize the 365 days ending on that purchase date. Python then assigns a segment using the latest active configuration in `DimModelConfig`.

This is a **purchase-triggered operational view**, not a table that continuously ages every inactive customer. Refreshing all customers after every transaction would be unnecessarily expensive for the team's intended near-real-time workflow. Month-end snapshots supply the complementary time-based view used for inactivity trends and historical comparison. Throughout the report, the word *live* refers to this operational purchase-time state; it should not be confused with the dashboard's final historical snapshot dated 30 June 2014.

# Chapter 4

# ETL Pipeline and Automation

## 4.1 The General View

Figure 4.1 summarizes the two complementary data paths and annual model training. The customer-demographics load and purchase-triggered RFM path support operational decisions; the monthly snapshot path preserves the state of customers at a common evaluation date.

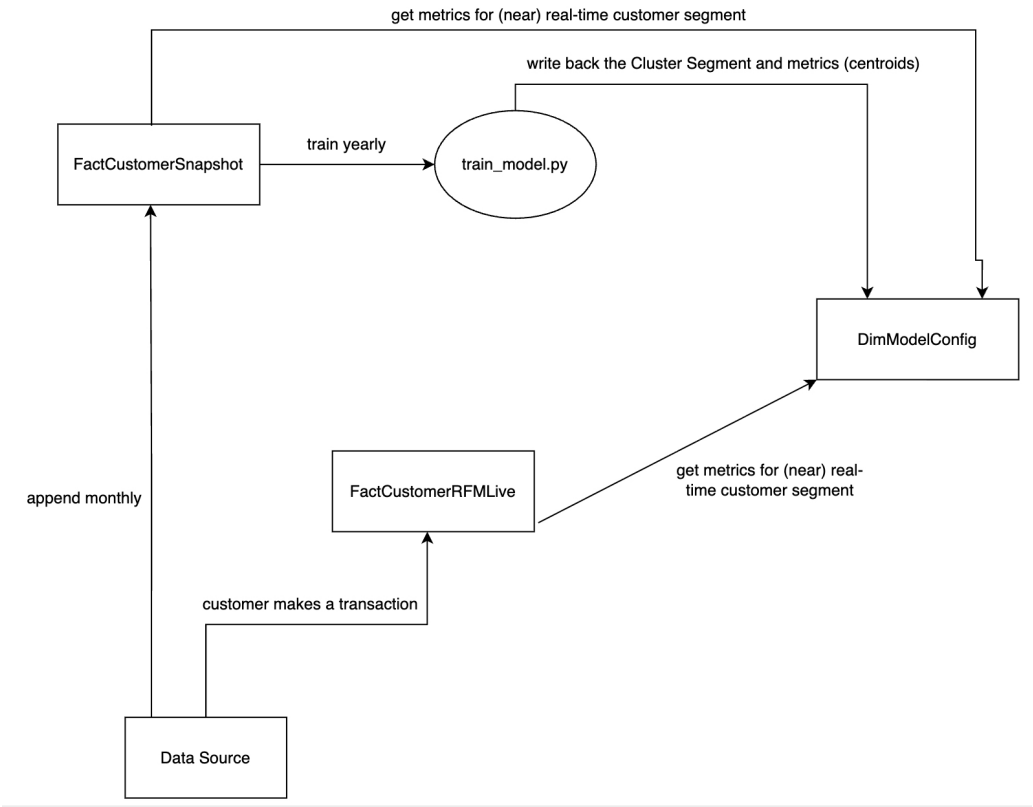



Figure 4.1: High-level flow from CompanyX transactions to RFM facts, model configurations, and reporting.

The first path begins with a customer transaction. The deployed one-minute SSIS jobs load changed customer profiles and refresh RFM for affected customers in **FactCustomerRFMLive**. The Python scorer then applies the latest active configuration from `DimModelConfig`. It log-transforms and standardizes the purchase-time RFM features, computes distances to stored centroids, and writes back the segment and model key. Because this view is evaluated on the most recent purchase date, a refreshed customer's recency is zero by design; customers are not rescored simply because a day passes.

The second path is scheduled for 00:05 on the first day of each month. It computes RFM as of the *previous month-end* directly from source transactions and appends the result to **FactCustomerSnapshot**. The historical snapshot scoring procedure uses stored model definitions. These snapshots are also the training source for new annual K-Means versions: training takes the year-end snapshot and writes scaler statistics, centroids, and segment mappings into `DimModelConfig` for subsequent scoring.

For the historical initial load, the team used `bootstrap.py` to create the first 2011 model and retrospectively label early snapshots once that model was available. Subsequent models were trained on the 2012 and 2013 year-end snapshots using `train_model.py`; no 2014 model was trained because source transactions end on 30 June 2014. This historical reconstruction must be distinguished from predictions genuinely available in those earlier months. The Power BI view discussed in Chapter 6 takes the 30 June 2014 historical snapshot as its final reference state.

## 4.2 ETL Pipeline & Data Flow

This section follows the three SSIS packages supplied in the project: `Package_Customer_Incremental.dtsx`, `RFM_Live.dtsx`, and `RFM_Snapshot.dtsx`. The following screenshots and SQL excerpts are from the team's implementation; full routine definitions are included in the system resources. The customer and live packages have been deployed on a one-minute schedule, while the snapshot package is scheduled monthly.

### 4.2.1 Customer Control Flow

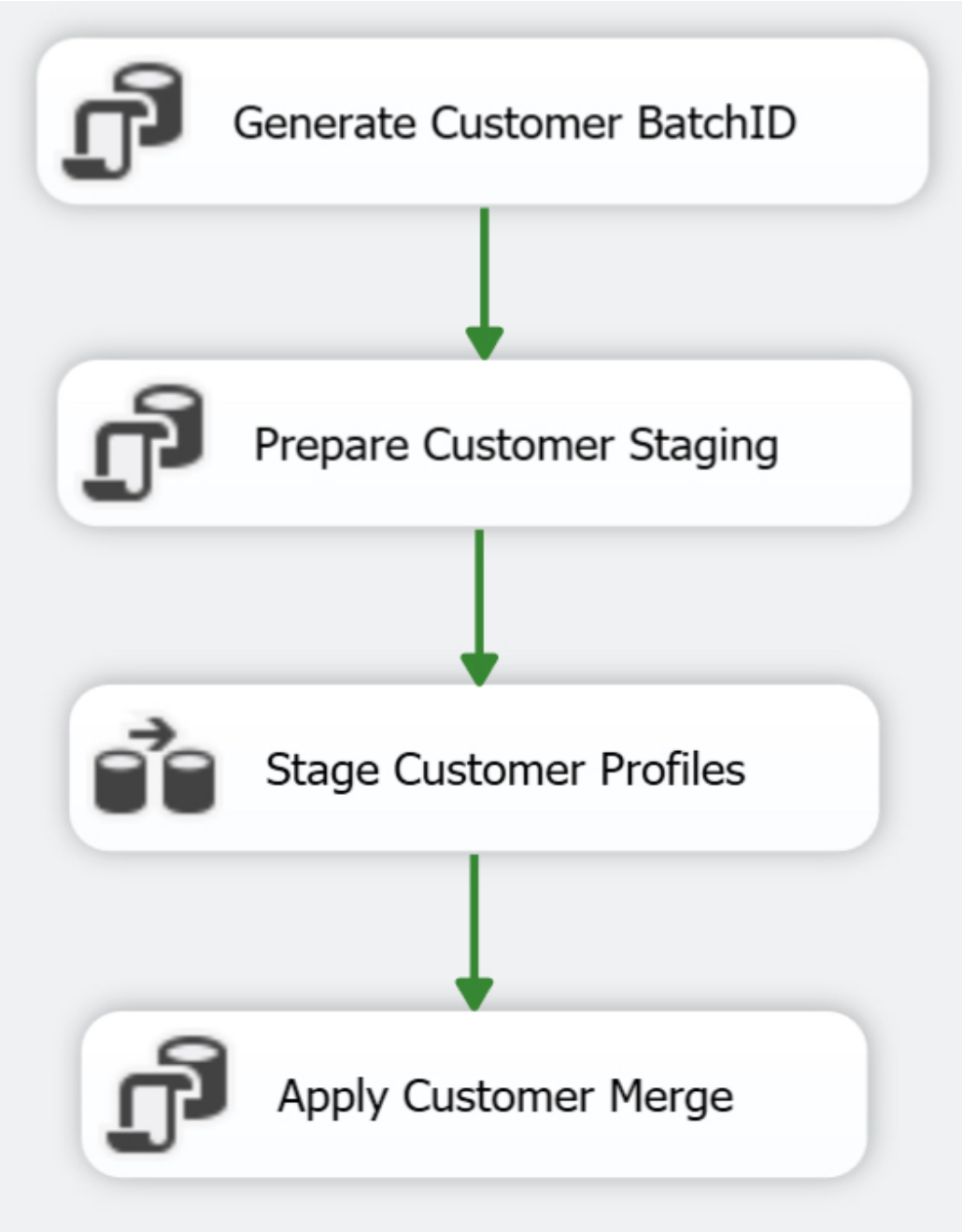



Figure 4.2: DimClusterProfiling and DimCustomerProfiling handling.

In the above figure, we show the overview of DimClusterProfiling and DimCustomerProfiling creation. First, a SQL task is executed to generate a new batch identifier:

```
SELECT CONVERT(VARCHAR(36), NEWID()) AS BatchID;
```

Listing 4.1: Batch ID Generation

After that, we clean the contents in previous runs in Preparing Customer Staging:

```
TRUNCATE TABLE CompanyX_DataWarehouse.stg.CustomerProfile;
```

Listing 4.2: Customer Profile Truncation

Following up, we examine the data flow inside Stage Customer Profiles:

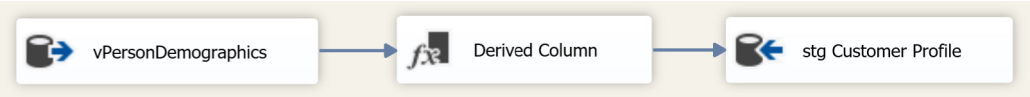



Figure 4.3: Data Flow of Stage Customer Profiles.

Overall, there are three components:

**nDemographics:** the OLE DB Source extracts changed demographic records from the CompanyX operational database:

```
SELECT
    c.CustomerID AS CustomerKey,
    d.BusinessEntityID,
    CAST(d.BirthDate AS DATE) AS BirthDate,
    d.Gender,
    d.MaritalStatus,
    d.YearlyIncome,
    d.TotalChildren,
    d.Education,
    d.Occupation,
    CAST(d.HomeOwnerFlag AS TINYINT) AS HomeOwnerFlag,
    d.NumberCarsOwned,
    CAST(d.DateFirstPurchase AS DATE) AS DateFirstPurchase,
    st.[Name] AS CountryName,
    st.[Group] AS RegionGroup
FROM Sales.vPersonDemographics AS d
INNER JOIN Sales.Customer AS c
    ON d.BusinessEntityID = c.PersonID
LEFT JOIN Sales.SalesTerritory AS st
    ON c.TerritoryID = st.TerritoryID
LEFT JOIN CompanyX_DataWarehouse.dbo.DimCustomerProfiling AS dc
    ON dc.CustomerKey_Str = CONVERT(NVARCHAR(20), c.CustomerID)
LEFT JOIN CompanyX_DataWarehouse.dbo.DimClusterProfiling AS dp
    ON dp.CustomerKey_Str = CONVERT(NVARCHAR(20), c.CustomerID)
   AND dp.IsCurrent = 1
WHERE d.DateFirstPurchase IS NOT NULL
```

```
27  AND (
28       dc.CustomerKey_Str IS NULL
29       OR dp.CustomerKey_Str IS NULL
30       OR EXISTS (
31            SELECT d.Gender, d.MaritalStatus, d.TotalChildren,
32                   CONVERT(BIT, d.HomeOwnerFlag), d.
    NumberCarsOwned
33            EXCEPT
34            SELECT dc.Gender, dc.MaritalStatus, dc.TotalChildren,
35                   dc.HomeOwnerFlag, dc.NumberCarsOwned
36       )
37       OR EXISTS (
38            SELECT d.YearlyIncome, d.Education, d.Occupation,
39                   st.[Name], st.[Group]
40            EXCEPT
41            SELECT dp.YearlyIncome, dp.Education, dp.Occupation,
42                   dp.CountryName, dp.RegionGroup
43       )
44  );
45
```

Listing 4.3: Table Selection

The extraction selects customers with a first purchase whose relevant demographic attributes are missing or different from the warehouse's current values. The `EXCEPT` predicates compare groups of attributes, including nullable values, without requiring a complete reload on every execution. This is a *change-detection* form of incremental loading rather than a source timestamp watermark.

**erived Column:** normalizes input values and handles missing demographic attributes before loading into staging. The configured fallback values include "Unknown, 0, and "U for selected attributes;

| Derived Column Name | Derived Column | Expression | Data Type | Length |
|---|---|---|---|---|
| CustomerKey_Str | <add as new column> | (DT_WSTR,20)CustomerKey | Unicode string [DT_WSTR] | 20 |
| BatchID | <add as new column> | (DT_GUID)("{" + @[User::BatchID] + "}") | unique identifier [DT_GUID] | |
| MaritalStatus | Replace 'MaritalStatus' | ISNULL(MaritalStatus) ? "U" : (UPPER(TRIM(MaritalStatus)) == "M" \|\| UPPER(TRIM(MaritalStatu... | Unicode string [DT_WSTR] | 1 |
| YearlyIncome | Replace 'YearlyIncome' | ISNULL(YearlyIncome) \|\| TRIM(YearlyIncome) == "" ? "Unknown" : TRIM(YearlyIncome) | Unicode string [DT_WSTR] | 30 |
| Gender | Replace 'Gender' | ISNULL(Gender) ? "U" : (UPPER(TRIM(Gender)) == "M" \|\| UPPER(TRIM(Gender)) == "F" ? UPP... | Unicode string [DT_WSTR] | 1 |
| CountryName | Replace 'CountryName' | ISNULL(CountryName) \|\| TRIM(CountryName) == "" ? "Unknown" : TRIM(CountryName) | Unicode string [DT_WSTR] | 50 |
| RegionGroup | Replace 'RegionGroup' | ISNULL(RegionGroup) \|\| TRIM(RegionGroup) == "" ? "Unknown" : TRIM(RegionGroup) | Unicode string [DT_WSTR] | 50 |
| TotalChildren | Replace 'TotalChildren' | ISNULL(TotalChildren) \|\| TotalChildren < 0 ? 0 : TotalChildren | four-byte signed integer [DT_I4] | |
| Education | Replace 'Education' | ISNULL(Education) \|\| TRIM(Education) == "" ? "Unknown" : TRIM(Education) | Unicode string [DT_WSTR] | 30 |
| Occupation | Replace 'Occupation' | ISNULL(Occupation) \|\| TRIM(Occupation) == "" ? "Unknown" : TRIM(Occupation) | Unicode string [DT_WSTR] | 30 |
| HomeOwnerFlag | Replace 'HomeOwnerF... | ISNULL(HomeOwnerFlag) ? (DT_UI1)0 : (HomeOwnerFlag ? (DT_UI1)1 : (DT_UI1)0) | Boolean [DT_BOOL] | |
| NumberCarsOwned | Replace 'NumberCarsO... | ISNULL(NumberCarsOwned) \|\| NumberCarsOwned < 0 ? 0 : NumberCarsOwned | four-byte signed integer [DT_I4] | |
| BirthDate | Replace 'BirthDate' | ISNULL(BirthDate) ? (DT_DBDATE)"1900-01-01" : BirthDate | database date [DT_DBDATE] | |

Figure 4.4: Derived Column: input data cleaning step.

these are data-cleaning conventions, not evidence that the original source value is known.

**stomer Profile:** the destination writes the transformed rows into the staging table described in Section 3.1, ready for the T-SQL merge.

Finally, the *Apply Customer Merge* SQL task executes the batch procedure if rows were staged. That procedure applies Type 0/1 customer rules and Type 2 demographic history in a transaction:

```
DECLARE @BatchID UNIQUEIDENTIFIER = CONVERT(UNIQUEIDENTIFIER, ?);
IF EXISTS (
    SELECT 1
    FROM stg.CustomerProfile
    WHERE BatchID = @BatchID
)
BEGIN
    EXEC dbo.usp_ApplyCustomerProfileBatch @BatchID = @BatchID;
END;
```

Listing 4.4: Merge

### 4.2.2 RFM Live

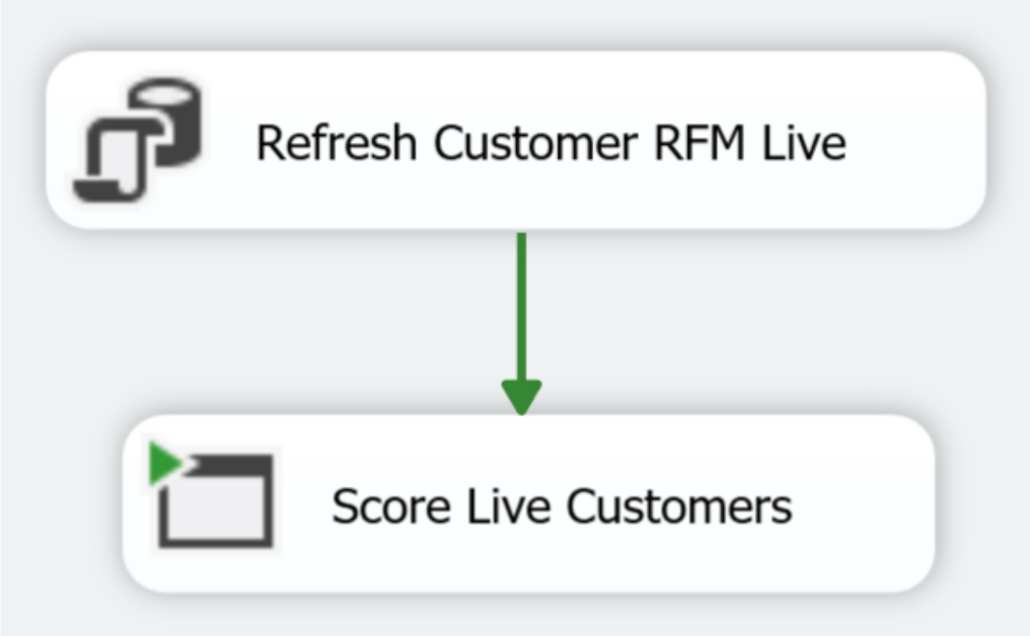



Figure 4.5: FactCustomerRFMLive handling.

The first SSIS task, *Refresh Customer RFM Live*, calls `usp_RefreshCustomerRFMLive`. A stored watermark tracks the last processed source modification timestamp. Changes to sales

headers or order details identify the affected customers; their full order history is aggregated once, using the customer's latest purchase date as the analysis date. The result is merged into `FactCustomerRFMLive` (insert for a new customer, update for a previously loaded customer).

The second, enabled task is *Score Live Customers*; its Python script is `live_clustering.py`. It loads the *latest active* model version, applies saved scaling parameters and centroids, and updates customers whose model key or segment has changed. An older SQL live-clustering task remains in the repository but is disabled in the deployed package and is not the operational scoring path. The approximately one-minute job interval makes this a *near-real-time batch* process rather than event-by-event synchronous processing.

### 4.2.3 RFM Snapshot

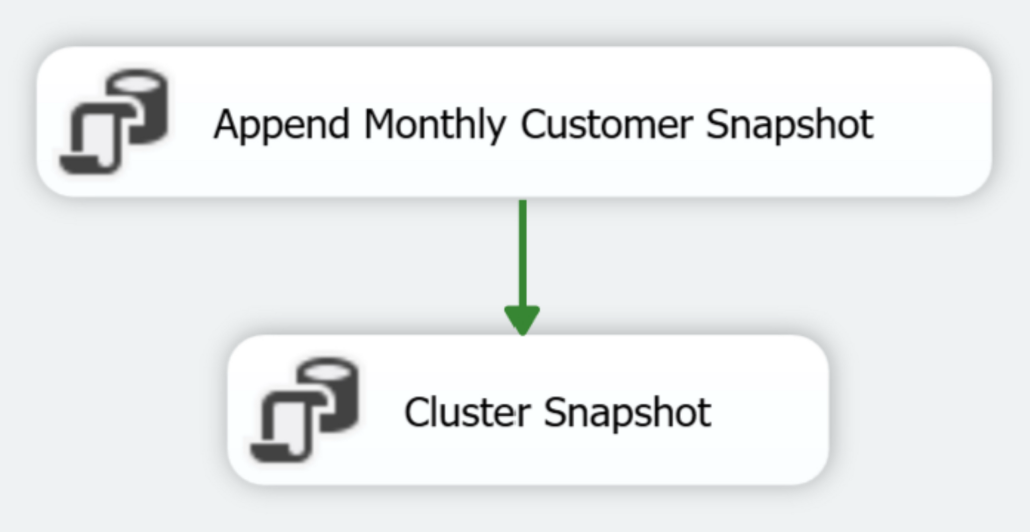



Figure 4.6: FactCustomerSnapshot handling.

The first task, *Append Monthly Customer Snapshot*, determines the previous month-end and calculates historical RFM values for the **FactCustomerSnapshot**:

```
DECLARE @SnapshotDate DATE;

SET @SnapshotDate = EOMONTH(GETDATE(), -1);

EXEC dbo.usp_AppendMonthlyRFMSnapshot
    @SnapshotDate = @SnapshotDate;
```

Listing 4.5: Monthly RFM Calculation

The subsequent *Cluster Snapshot* task assigns segment labels using the available effective-dated model definitions:

```
EXEC dbo.usp_ClusterCustomerRFMSnapshot;
```

Listing 4.6: Monthly Cluster Segment

## 4.3 Annual Model Training and Scheduling

The training scripts use the final snapshot of a year as their input. The first model was established by `bootstrap.py` using the 2011 year-end data, with historical 2011 snapshots labeled retrospectively. The next training runs were executed for 2012 and 2013; each saved model version has five cluster rows and a validity start date in `DimModelConfig`. The active live scorer uses the most recently active model, while the historical snapshot procedure selects effective-dated model configurations and includes bootstrap handling for the earliest periods.

The project includes the Airflow DAG `rfm_year_end_training`, which specifies a yearly run on 1 January. At the time of submission, the DAG's unattended end-to-end execution had not been verified. In particular, its configured start time (00:00) precedes the monthly SSIS snapshot schedule (00:05); a production-ready dependency would require waiting for the December snapshot to be committed before training begins. The DAG also needs an environment-independent Python module path on the deployment machine. These do not prevent the separately executed annual training workflow from being demonstrated, but remain operational integration tasks.

## 4.4 Pipeline Outputs

Successful runs produce one current Type 1 customer profile per customer, effective-dated Type 2 demographic records, a purchase-triggered RFM record for affected customers, periodic monthly customer snapshots, and a persisted set of cluster parameters. The Power BI dashboard reads the warehouse's historical June 2014 analytical view. This separation allows both immediate post-purchase decisions and longer-term segment comparisons without requiring an update of every live customer after each purchase.

# Chapter 5

# RFM Analysis and K-Means Clustering

## 5.1 Motivation

The CompanyX transactional database contains purchases from May 2011 through June 2014. However, raw sales records alone do not identify which customer groups require different business strategies. This chapter follows the team's implemented approach: summarize purchase behaviour using RFM, clean and transform the features, fit K-Means to customer snapshots, and translate the resulting numeric clusters into interpretable segment names. The trained configurations can then be reused by the operational scoring pipeline and historical snapshot process.

### 5.1.1 RFM Variables

Recency (R) is the number of days from an evaluation date to the customer's last purchase. Frequency (F) is the count of distinct sales orders within the trailing 365 days. Monetary (M) is the sum of order-line amounts within that same 365-day period. Recency may exceed 365 days for customers who have not purchased recently. The year-end training snapshots used by the project are 31 December 2011, 2012, and 2013; there is no 2014 trained model.

## 5.2 Data Cleaning

The raw data is first ingested from the FactCustomerSnapshot, and then processed through deduplication. After that, any R, F, or M values are missing, the process stops and reports the number of affected rows to the ETL Team. Anomalies values such as negative values are set to zero, since a purchase count or a monetary value can not be negative.

This subset of data serves a representative overview of customer purchases based on RFM analysis as mentioned above.

| Customer | Recency | Frequency | Monetary |
|---|---|---|---|
| 11057 | 37 | 1 | 3729.36 |
| 11058 | 22 | 1 | 3756.99 |
| 11060 | 12 | 1 | 3756.99 |
| 11061 | 18 | 1 | 3729.36 |
| 11072 | 0 | 1 | 3756.99 |
| 11238 | 84 | 1 | 3756.99 |
| 11239 | 71 | 1 | 3729.36 |
| 11240 | 32 | 1 | 3756.99 |
| 11243 | 38 | 1 | 3756.99 |
| 11244 | 47 | 1 | 3756.99 |
| 11247 | 41 | 1 | 3729.36 |
| 11250 | 14 | 1 | 3756.99 |
| 11333 | 10 | 1 | 3756.99 |
| 11334 | 27 | 1 | 3756.99 |
| 11337 | 22 | 1 | 3729.36 |
| 11591 | 79 | 1 | 3953.99 |
| 11592 | 79 | 1 | 3953.99 |
| 11593 | 69 | 1 | 3953.99 |
| 11599 | 72 | 1 | 3953.99 |
| 11601 | 71 | 1 | 3953.99 |
| 11606 | 82 | 1 | 3953.99 |
| 11607 | 71 | 1 | 3953.99 |

Table 5.1: Example RFM records before preprocessing.

## 5.3 Log transformation

Spending and purchase counts are usually right-skewed. Most customers spend a moderate amount, while only a small number spend many times more. If raw values are used, these large values stretch the scale and the K-Means model spends more effort unnecessarily separating a few high spenders (outliers) rather than grouping typical customers. By using log transformation, the issue can be minimized.

For example, a customer who spends 3,000 and 300,000 differs by a factor of 100 in raw terms, but their log values differ by only 4.6. We use log(1+x) in this case, written log1p because some customers have a value of zero, which leads to log(0) undefined.

| Customer | Recency | Frequency | Monetary |
|---|---|---|---|
| 11057 | 3.638 | 0.693 | 8.224 |
| 11058 | 3.135 | 0.693 | 8.232 |
| 11060 | 2.565 | 0.693 | 8.232 |
| 11061 | 2.944 | 0.693 | 8.224 |
| 11072 | 0.000 | 0.693 | 8.232 |
| 11238 | 4.443 | 0.693 | 8.232 |
| 11239 | 4.277 | 0.693 | 8.224 |
| 11240 | 3.497 | 0.693 | 8.232 |
| 11243 | 3.664 | 0.693 | 8.232 |
| 11244 | 3.871 | 0.693 | 8.232 |
| 11247 | 3.738 | 0.693 | 8.224 |
| 11250 | 2.708 | 0.693 | 8.232 |
| 11333 | 2.398 | 0.693 | 8.232 |
| 11334 | 3.332 | 0.693 | 8.232 |
| 11337 | 3.135 | 0.693 | 8.224 |
| 11591 | 4.382 | 0.693 | 8.283 |
| 11592 | 4.382 | 0.693 | 8.283 |
| 11593 | 4.248 | 0.693 | 8.283 |
| 11599 | 4.290 | 0.693 | 8.283 |
| 11601 | 4.277 | 0.693 | 8.283 |
| 11606 | 4.419 | 0.693 | 8.283 |
| 11607 | 4.277 | 0.693 | 8.283 |

Table 5.2: Log-transformed sample RFM values.

## 5.4 Standardization

Before importing data to run the K-Means Clustering model, it is essential to standardize the RFM values to ensure that each metric contributes equally to the analysis, since Recency, Frequency, and Monetary values are measured on different scales. K-Means relies heavily on distances (later will be shown), so a variable with a larger spread would dominate the result.

StandardScaler solves this by subtracting the mean and dividing by the standard deviation of each variable, so that each one has a mean of 0 and a standard deviation of 1. The mean and standard deviation are estimated from the training data and stored, so that new customers are scaled in the same way.

| Customer | Recency | Frequency | Monetary |
|---|---|---|---|
| 11057 | 0.109 | 0.000 | -0.875 |
| 11058 | -0.392 | 0.000 | -0.581 |
| 11060 | -0.962 | 0.000 | -0.581 |
| 11061 | -0.583 | 0.000 | -0.875 |
| 11072 | -3.522 | 0.000 | -0.581 |
| 11238 | 0.913 | 0.000 | -0.581 |
| 11239 | 0.747 | 0.000 | -0.875 |
| 11240 | -0.032 | 0.000 | -0.581 |
| 11243 | 0.135 | 0.000 | -0.581 |
| 11244 | 0.342 | 0.000 | -0.581 |
| 11247 | 0.209 | 0.000 | -0.875 |
| 11250 | -0.819 | 0.000 | -0.581 |
| 11333 | -1.128 | 0.000 | -0.581 |
| 11334 | -0.196 | 0.000 | -0.581 |
| 11337 | -0.392 | 0.000 | -0.875 |
| 11591 | 0.852 | 0.000 | 1.454 |
| 11592 | 0.852 | 0.000 | 1.454 |
| 11593 | 0.719 | 0.000 | 1.454 |
| 11599 | 0.761 | 0.000 | 1.454 |
| 11601 | 0.747 | 0.000 | 1.454 |
| 11606 | 0.889 | 0.000 | 1.454 |
| 11607 | 0.747 | 0.000 | 1.454 |

Table 5.3: Standardized sample RFM values.

## 5.5 K-Means Clustering

K-Means is an unsupervised algorithm that partitions observations into $K$ clusters by iteratively assigning each point to the nearest centroid and updating centroid locations. The project fits scikit-learn `KMeans` with $K = 5$, `k-means++` initialization, `n_init=10`, `max_iter=300`, and `random_state=42`.

For each data point in the dataset, calculate the Euclidean distance between the point and each centroid. Assign the data point to the cluster with the closest centroid

$$d(\mathbf{x}, \mathbf{c}) = \sqrt{\sum_{j=1}^{3}(x_j - c_j)^2}.$$

Where:

- $d(x, y)$: the distance between data point x and centroid y
- $y_i$: donates the value of the i-th attribute value of centroid y
- $x_i$: represents the i-th attribute of the data point x
- $i$ : refers to index of the attribute
- $n$ : indicates the total number of the attribute for each data point

Within-cluster sum of squares (WCSS), also called inertia, measures the total squared distance from observations to their assigned centroids:

$$\text{WCSS} = \sum_{k=1}^{K}\sum_{i \in S_k}\sum_{j=1}^{3}(x_{ij} - c_{kj})^2.$$

Where:

- K: Total number of clusters.
- k: An index that refers to the current cluster, ranging from 1 to K.
- $S_k$: The set of data points assigned to cluster k.
- p: Number of features or dimensions in the dataset.
- j: An index that refers to the current feature or dimension, ranging from 1 to p.
- $x_{ij}$: The j-th feature value of the i-th data point.
- $\bar{x}_{kj}$: The mean value of the j-th feature in cluster k (i.e., the centroid's value).

- $(x_{ij} - \bar{x}_{kj})^2$: The squared Euclidean distance between the j-th feature of data point xi and the corresponding feature of the centroid of its cluster.

This WCSS can also be known as the Inertia, the lower means the tighter the data points are grouped. Another important criteria, which is Silhouette, also needs to be considered. The silhouette score measures how well-separated and cohesive clusters are in K-Means clustering. It evaluates each data point's fit relative to its own cluster versus neighboring clusters.

Figures 5.1 and 5.2 reproduce the diagnostic plots already saved with the project for the 31 December 2013 snapshot. Inertia decreases as $K$ increases, while the silhouette metric measures separation between clusters. The silhouette score for $K = 5$ is approximately 0.518; $K = 2$ achieves a higher score (approximately 0.636) on this snapshot. Therefore five clusters were chosen as a *business-interpretability trade-off*, supported by the elbow/silhouette diagnostics and the desire for distinct actionable customer profiles, not because five globally maximizes the silhouette score.

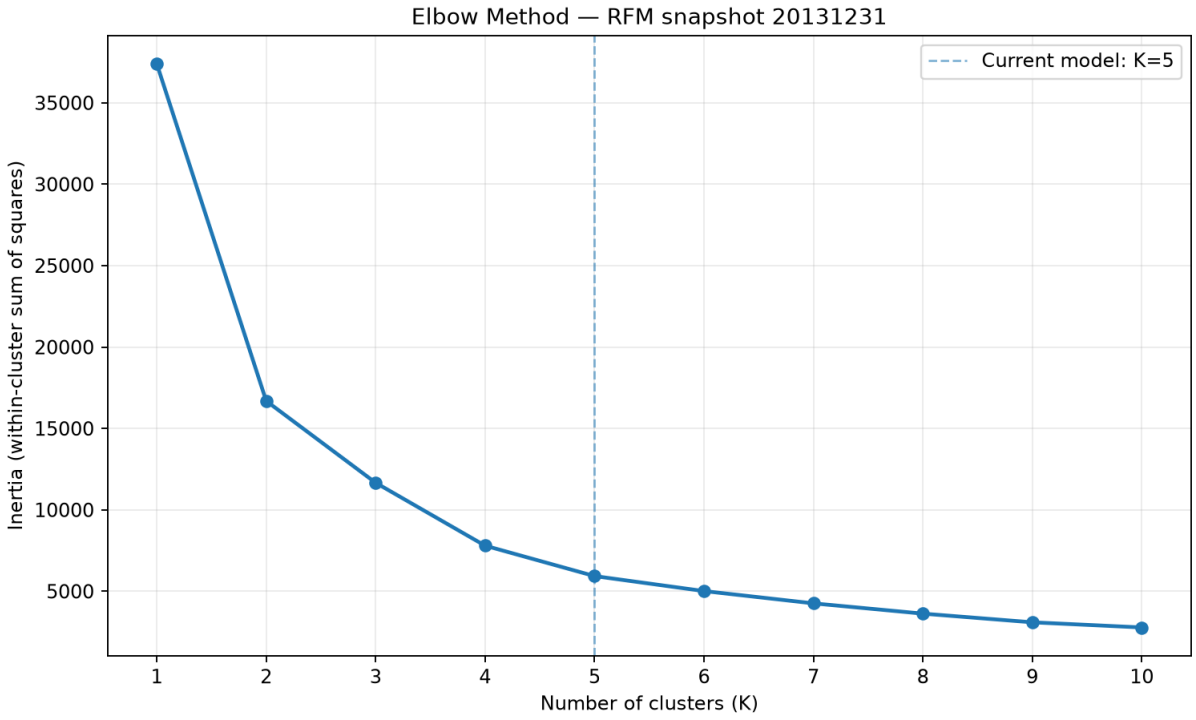



Figure 5.1: Elbow plot for the 31 December 2013 RFM snapshot (existing project diagnostic).

K-Means only returns a numeric cluster ID (0 to 4) for each customer, which reflects no value of what the groups mean. Moreover, annually the model is retrained and the number from 0 to 4 is different and not staying consistent. Therefore, to turn them into meaningful clusters automatically, a solution is implemented: Each cluster on an RFM variable is given "points", based on how its average compares with the other clusters. The rules are

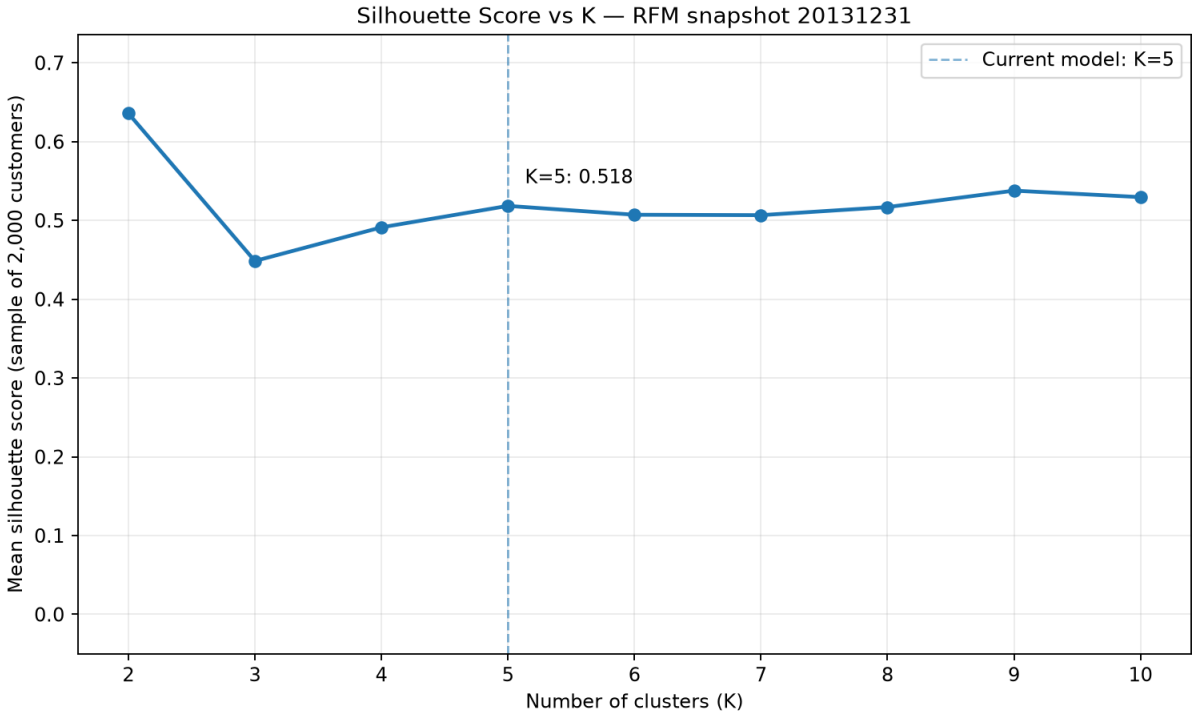



Figure 5.2: Silhouette comparison for different cluster counts on the same snapshot.

For Monetary and Frequency: the cluster with the highest average gets 5 points and the lowest get 1. (the higher these values are, the more valuable they are)

For Recency: the rule is reversed because shorter time since the last purchase is better. The implemented composite score is Monetary rank + Frequency rank + 1.1 × Recency rank.

Recency is weighted slightly higher, because a recent purchase is a strong sign that a customer is still active. Points are given to clusters, not to individual customers.

The outcome will look like this:

| Cluster | Avg Recency | Avg Frequency | Avg Monetary | R points | F points | M points | Score | Segment |
|---|---|---|---|---|---|---|---|---|
| 4 | 110.34 | 2.55 | 1,694.00 | 4 | 5 | 5 | 14.4 | Champions |
| 2 | 13.31 | 1.24 | 96.10 | 5 | 4 | 3 | 12.5 | Recent / Promising |
| 0 | 174.17 | 1.00 | 1,518.38 | 3 | 2 | 4 | 9.3 | High-Value / Cooling Off |
| 3 | 196.03 | 1.03 | 48.91 | 2 | 3 | 2 | 7.2 | Low-Spend / Occasional |
| 1 | 602.92 | 0.00 | 0.00 | 1 | 1 | 1 | 3.1 | Lost / Dormant |

The example model output was recorded for an 18,484-customer analysis extract. The final dashboard chapter reports 18,485 customers for its June 2014 view. This one-record difference has not been reconciled in the submitted materials, so the two counts should not be presented as an identical population without verification.

Champions (cluster 4) has the highest frequency and spend, with an average of 2.55 purchases and 1,694 in total. Its average recency is 110 days, so these customers are valuable but have not bought recently.

High-Value / Cooling Off (cluster 0) is the second-largest group by spend, but each customer has bought only once on average. Their last purchase was about 174 days ago, so they are at risk of leaving.

Low-Spend / Occasional (cluster 3) is the largest group. Customers buy about once, spend about 49 on average, and last bought about 196 days ago.

Recent / Promising (cluster 2) bought most recently, with an average of 13 days since the last purchase. Their spend is low so far, which suggests they are new customers who may grow.

Lost / Dormant (cluster 1) has no recorded frequency or spend in the period, and an average recency of 603 days. These customers have stopped buying.

## 5.6 Conclusion

The result of the RFM Analysis and K-Means clustering reveal the distinct customer segments with unique purchasing behaviours. By standardizing the RFM variables and applying the K-Means algorithm, the dataset was successfully divided into five interpretable clusters. Such segmentation highlights not only the diversity in customer engagement levels but also provides valuable insights for strategic business decision-making. Therefore, the CompanyX will be able to design and launch effective marketing strategies to each cluster of customers, not only to retain the occasional buyers, but also to come up with attractive promotions for recent customers.

The first historical model was initialized with `bootstrap.py` from the 2011 year-end snapshot, including retrospective labels for earlier months. Separate 2012 and 2013 models were then saved with `train_model.py`. The most recent active model is used by the Python live scorer. These saved model versions support repeatable clustering without retraining on every purchase.

# Chapter 6

# Business Intelligence & Analytics Dashboards

## 6.1 Context & Dashboard Objectives

### 6.1.1 Business Context

After completing the data warehouse and running the K-Means clustering model, our team needed an interactive tool to show the results to decision-makers. In the database, customer data and cluster numbers are stored in tables, which are difficult for managers to read and understand quickly. Therefore, we built a Business Intelligence (BI) dashboard using Microsoft Power BI.

The dashboard uses the current warehouse, but its *live analytical view for reporting* is the `FactCustomerSnapshot` slice at `SnapshotDateKey = 20140630`, rather than the physical `FactCustomerRFMLive` table. This matters because the latter deliberately contains purchase-time recency of zero for refreshed customers, while a common month-end date is needed to compare inactivity. At this stage, the dashboard is integrated but Page 1 remains subject to final team refinement. The three pages focus on:

1. **Executive Overview:** Help company directors monitor total sales, customer numbers, yearly trends, and early warnings about customer churn.

2. **Customer Demographics:** Help marketing and sales teams understand customer profiles, such as location, income level, age group, gender, and job types.

3. **RFM Segmentation and Strategy:** Help the CRM team see how each customer cluster behaves in terms of Recency, Frequency, and Monetary value, and choose the right marketing strategy for each group.

### 6.1.2 Visualization Principles

To ensure that the dashboards are clear, consistent, and easy to interpret, we applied data visualization principles discussed in the course and in Wilke (2019). The selection of chart types was based on the characteristics of the data and the analytical purpose of each visualization.

- **Treemaps:** Used to represent categorical distributions, such as customer occupations, without overcrowding the visualization.
- **100% Stacked Column Charts:** Used to compare the relative proportions of customer segments across different time periods, highlighting changes in segment composition.
- **Age Distribution Charts:** Customers are grouped into age ranges to provide a clearer overview of the demographic distribution.
- **Bubble Charts:** Used to visualize the relationship between Recency and Monetary values, with bubble size representing the number of customers in each segment.
- **Consistent Segment Colors:** Each customer segment is assigned a fixed color across all three dashboard pages, allowing users to identify and compare segments more easily. The colors are defined as follows:
  - **Champions:** Red (#A82020)
  - **High-Value / Cooling Off:** Dark Green (#22593D)
  - **Recent / Promising:** Orange (#D97F1A)
  - **Low-Spend / Occasional:** Teal (#259696)
  - **Lost / Dormant:** Purple (#4C2773)

### 6.1.3 Dashboard Structure

The Power BI report consists of three pages, each serving a different analytical purpose:

1. **Page 1 – Executive Customer & Segmentation Overview:** Presents key performance indicators and trends in customer activity and segment distribution.
2. **Page 2 – Customer Demographics & Persona Profiling:** Examines demographic characteristics and allows users to explore the profiles of individual customer segments.
3. **Page 3 – Customer RFM Segmentation & Behavior:** Compares the purchasing behavior of customer segments through RFM metrics and visualizations.

## 6.2 Page 1: Executive Customer & Segmentation Overview

Page 1 is designed for high-level managers who need a quick summary of the company's customer base, revenue, and retention status.

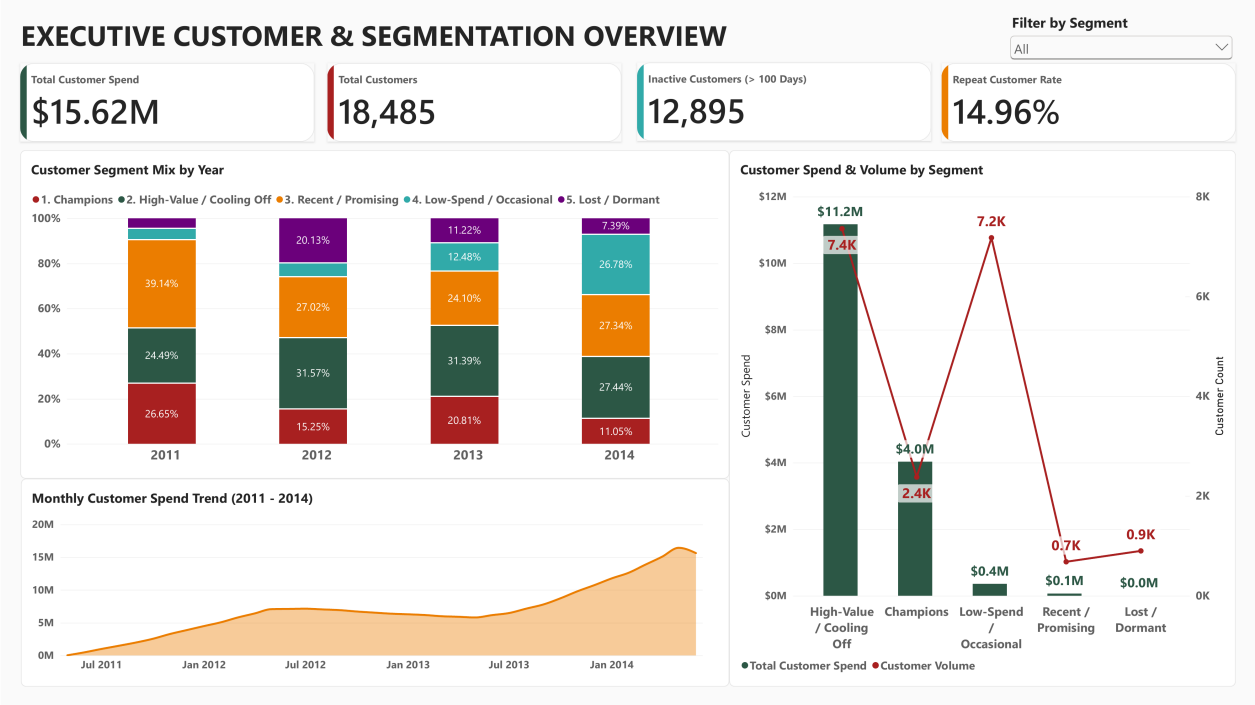



Figure 6.1: Executive Customer & Segmentation Overview Dashboard (Page 1).

### 6.2.1 Core Executive KPIs

At the top of Page 1, we place four key metrics:

1. **Total Customer Spend ($15.62M):** Sum of the 365-day customer monetary values in the dashboard reference snapshot ($15,615,354.05), rather than verified lifetime revenue.

2. **Total Customers (18,485):** Distinct customers in the imported 30 June 2014 reporting view; this is not a claim about all possible customers in the source database.

3. **Inactive Customers (> 100 Days) (12,895):** Customers whose snapshot recency exceeds 100 days as of 30 June 2014. This makes up **69.76%** of all customers. In the bicycle retail industry, bikes are durable goods, so people do not purchase every month. However, after 100 days (more than 3 months), if a buyer does not come back for small accessories or maintenance, they are at risk of leaving. Therefore, this KPI serves as an early-warning signal.

4. **Repeat Customer Rate (14.96%):** Calculated as the percentage of customers who have ordered at least two times:

$$\text{Repeat Customer Rate} = \frac{\sum[\text{Customers with Frequency} \geq 2]}{\text{Total Customers}} = \frac{2{,}765}{18{,}485} = 14.96\% \quad (6.1)$$

This indicates that 85.04% do not have at least two orders *within the trailing 365-day measurement window*. It does not establish that these customers purchased only once over their entire history.

### 6.2.2 Visual Charts and Key Findings

**1. Customer Segment Mix by Year (2011–2014)**

We selected a 100% Stacked Column Chart to show the proportion of each customer segment over the four operational years:

- **2011 Baseline:** In the first year, CompanyX focused on establishing its retail presence. *Recent / Promising* customers constituted the largest group at **39.14%**, followed by *Champions* at **26.65%** and *High-Value / Cooling Off* at **24.49%**. The *Low-Spend / Occasional* and *Lost / Dormant* segments formed only a small portion (roughly 9.7% combined).

- **2012 Transition:** As earlier buyers matured, the proportion of *High-Value / Cooling Off* customers increased to **31.57%**. However, buyers who did not place repeat orders transitioned into *Lost / Dormant*, which rose to **20.13%**. The share of *Champions* stood at **15.25%**, while *Recent / Promising* made up **27.02%**.

- **2013 Product Diversification:** In 2013, the business began offering bike accessories and apparel. As a result, the *Low-Spend / Occasional* segment grew substantially to **12.48%**. The *High-Value / Cooling Off* group remained steady at **31.39%**, and *Champions* accounted for **20.81%**.

- **2014 Mass Market Expansion:** By 2014, the proportion of *Low-Spend / Occasional* shoppers more than doubled to **26.78%**. Meanwhile, *High-Value / Cooling Off* represented **27.44%**, *Recent / Promising* was **27.34%**, and *Champions* made up **11.05%**. The *Lost / Dormant* group dropped to **7.39%**.

These figures illustrate an important strategic shift: while CompanyX was originally a niche retailer of premium bicycles dominated by high-value buyers in 2011–2012, it successfully expanded to general everyday shoppers in 2013–2014. Nevertheless, this expansion brought a large volume of one-time or low-spending buyers.

**2. Rolling Customer Spend at Monthly Snapshots (2011–2014)**

The Area Chart follows the sum of trailing-365-day customer monetary values at successive monthly snapshot dates (July 2011 to June 2014). It shows a *rolling annual spending measure sampled monthly*, not independent sales revenue earned within each calendar month:

- **Early Stage (2011–Mid 2012):** The rolling spend measure started at a moderate level, which was typical for a new business with a limited product catalog.
- **Rapid Acceleration (Mid 2012–End 2013):** Starting in the second half of 2012, rolling customer monetary totals increased significantly. The chart shows strong upward momentum throughout 2013, reaching a high rolling monetary value towards the end of 2013.
- **Seasonality Trends:** Changes in the rolling total may reflect both newly included purchases and older purchases leaving the 365-day window. A claim about monthly seasonality would require a separate measure of sales booked within each individual month.

**3. Customer Spend & Volume by Segment**

This Dual-Axis Combo Chart displays total sales as bars on the left axis and customer counts as points on the right axis:

- **Spending Concentration (rather than a strict 80/20 split):**
  - The *High-Value / Cooling Off* cluster generates **$11.2M** (71.53% of total revenue) from 7,357 customers (39.80% of the customer base).
  - The *Champions* cluster contributes **$4.0M** (25.81% of total revenue) from 2,377 customers (12.86%).
  - Together, these two premium segments contribute **97.34%** of the snapshot's rolling-365-day monetary total ($15.2M out of $15.62M), while making up only **52.66%** of all customers (9,734 people).
- **The Low-Monetary Contrast:** In comparison, the *Low-Spend / Occasional* group contains 7,175 customers (38.82% of the total base) but brings in only **$0.35M** (2.25% of revenue). Similarly, *Recent / Promising* accounts for **$0.07M** (0.42%), and *Lost / Dormant* accounts for **$0.00**.
- **Key Takeaway:** These figures motivate prioritizing high-value retention campaigns; whether a campaign is more profitable than acquisition must be assessed separately using cost and response data.

### 6.2.3 Dashboard Interactivity and Slicer Architecture

In our Power BI solution, an interactive dropdown slicer (*Filter by Segment*) is placed in the top-right corner of each dashboard page. However, during the dashboard development, an important architectural decision was required for Page 1:

- **The Macro Trend Conflict:** The two longitudinal visuals—*Customer Segment Mix by Year* and *Rolling Customer Spend Trend*—rely on historical snapshot data from the periodic snapshot table (`FactCustomerSnapshot`) to show the whole company's evolution. In particular, the 100% stacked column chart displays the structural proportions among all five segments. If a single segment is filtered, the multi-segment composition collapses to 100%, eliminating its analytical purpose.
- **Implementation of Edit Interactions:** Following standard Business Intelligence best practices, we accessed the *Format* → *Edit Interactions* menu in Power BI and set the interaction from the segment slicer to **None** for these two macro trend visuals. As a result, users can filter the top KPI cards and the segment combo chart without breaking the overall company baseline timeline.

## 6.3 Page 2: Customer Demographics & Persona Profiling

Page 2 helps us understand who our customers are, where they live, and what their personal characteristics look like.

### 6.3.1 Demographic Overview KPIs

The four cards on the left show customer averages:

- **Total Customers:** 18,485.
- **Total Customer Spend:** $15.62M.
- **Average Spend per Customer: $844.76**.
- **Average Order Value (AOV): $720.53**.

These measures describe the same trailing 365-day period, and their ratio is consistent with comparatively low order frequency. They do not by themselves establish customers' lifetime order counts.

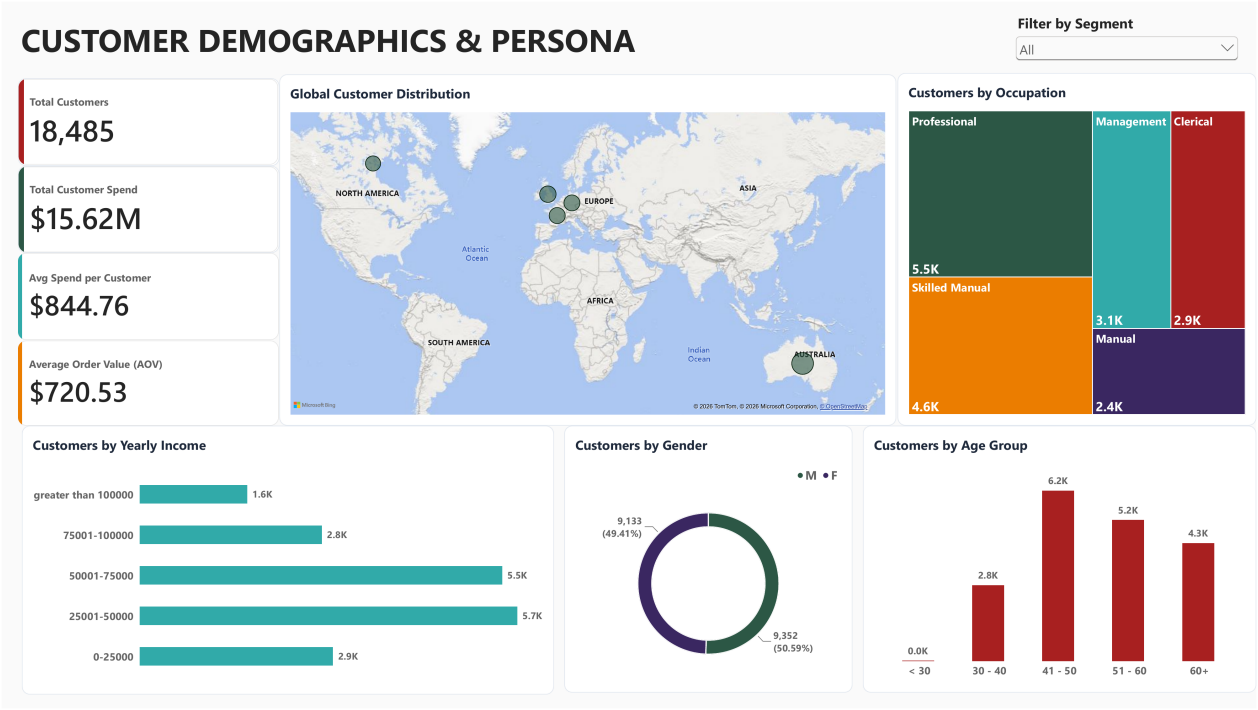



Figure 6.2: Customer Demographics & Persona Dashboard (Page 2).

### 6.3.2 Demographic Analysis (Overall Base)

#### 1. Geographic Distribution (Map Visual)

The map shows that our customers live in developed markets with strong cycling cultures, mainly in North America (United States, Canada), Western Europe (United Kingdom, Germany, France), and Australia.

#### 2. Customers by Occupation (Treemap)

- **Professional** (5,505 people, 29.8%) and **Skilled Manual** (4,577 people, 24.8%) make up more than **54.6%** of all customers.
- Other occupations include Management (3,075, 16.6%), Clerical (2,928, 15.8%), and Manual workers (2,400, 13.0%).
- This shows that our core buyers are educated workers and skilled technicians who have good income and an active lifestyle.

#### 3. Customers by Yearly Income (Horizontal Bar Chart)

- The middle-income groups are the largest: **$25,001–$50,000** has 5,720 customers (30.9%), and **$50,001–$75,000** has 5,496 customers (29.7%).

- Together, middle-class customers make up **60.6%** of the total customer base. High earners ($> \$100,000$) represent 1,600 customers (8.7%), while entry-level earners ($0–$25,000) account for 2,870 customers (15.5%).

**4. Customers by Gender (Donut Chart)**

- **Male:** 9,352 customers (50.59%).
- **Female:** 9,133 customers (49.41%).
- The gender ratio is almost 50/50, suggesting a balanced gender composition in the analyzed customer population.

**5. Customers by Age Group (Column Histogram)**

- Customers are grouped into age brackets: $< 30$, $30 - 40$, $41 - 50$, $51 - 60$, and 60+.
- The majority of customers are in middle age: **41–50 years old** (6,200 people, 33.5%) and **51–60 years old** (5,200 people, 28.1%).
- People under 30 make up a very small portion, within the age distribution captured by the dashboard; product-price preferences cannot be inferred from age alone.

### 6.3.3 Segment-Specific Demographic Personas (Slicer Drill-Down)

By utilizing the interactive segment slicer on Page 2, decision-makers can drill down into the unique demographic persona of each customer cluster. Table 6.1 provides an empirical comparison across the five groups.

Table 6.1: Comparative Demographic Profiles Across Customer Segments.

| Segment | Customers | Total Spend | AOV ($) | Top Occupation | Peak Income | Dominant Age |
|---|---|---|---|---|---|---|
| 1. Champions | 2,377 | $4.03M | $665.99 | Professional (39.2%) | $50k–$75k (30.9%) | 41–50 (38.3%) |
| 2. High-Value | 7,357 | $11.17M | $1,518.20 | Professional (29.9%) | $50k–$75k (32.2%) | 41–50 (39.4%) |
| 3. Recent | 679 | $65.18K | $77.22 | Skilled Manual (28.7%) | $25k–$50k (29.8%) | 60+ (29.5%) |
| 4. Low-Spend | 7,175 | $350.92K | $47.33 | Professional (27.9%) | $25k–$50k (30.6%) | 60+ (29.3%) |
| 5. Lost / Dormant | 897 | $0.00 | $0.00 | Skilled Manual (37.8%) | $25k–$50k (43.5%) | 41–50 (39.6%) |

**Key Findings from Segment Drill-Down**

Filtering each segment on Page 2 highlights significant differences in purchasing power, age distribution, and professional background:

1. **1. Champions (Affluent White-Collar Repeat Buyers):**

- *Highest Earning Power:* Champions have the highest income level among all clusters. Buyers earning more than $100,000 account for **14.01%** (333 individuals), while 30.88% earn $50,000–$75,000.
- *White-Collar Roles: Professional* (39.17%, 931 people) and *Management* (18.93%, 450 people) represent **58.10%** of this group, demonstrating that corporate earners drive repeat purchases.
- *Distinct Repeat Behavior:* This is the only segment where Average Spend per Customer ($1,700) is much higher than Average Order Value ($665.99). This supports the conclusion that Champions place repeat orders ($F \approx 2.55$). Product-specific upgrades or replacements would need an order-category analysis.
- *Age Concentration:* Heavily centered in the **41–50** age bracket (**38.33%**, 911 buyers), followed by 51–60 (27.93%). These are established professionals investing in active health and cycling hobbies.

2. **2. High-Value / Cooling Off (Single-Purchase Premium Buyers):**

- *Spend Equals AOV:* Average Spend per Customer exactly equals Average Order Value at **$1,518.20**. This is consistent with approximately one purchase in the measured 365-day window; the RFM totals alone cannot establish the precise product purchased or lifetime ordering behaviour.
- *Solid Middle Class:* The largest income groups are $50,001–$75,000 (32.17%) and $25,001–$50,000 (31.58%), representing **63.75%** of the cluster.
- *Age Group:* The dominant bracket is 41–50 years old (39.42%, 2.9K buyers), followed by 51–60 (28.55%, 2.1K buyers).

3. **3. Recent / Promising (Older Entrants & Skilled Technicians):**

- *Unexpected Age Shift:* In contrast to Champions and High-Value clusters, the single largest age group in *Recent / Promising* is **60+ years old** (**29.46%**, 200 customers). This demonstrates that senior riders and retirees represent a growing acquisition channel.
- *Occupation Distribution: Skilled Manual* (28.72%, 195 people) surpasses *Professional* (27.54%, 187 people) as the leading occupation.
- *Early Repeat Signals:* With an AOV of $77.22 and an average spend of $95.99, a subset of customers have already completed second accessory orders within two weeks ($R = 13.28$ days).

4. **4. Low-Spend / Occasional (Budget-Conscious Accessory Shoppers):**

- *Lower-Income Skew:* This group contains the highest percentage of lower-income shoppers: buyers earning below $25,000 account for **20.60%** (1,478 individuals).
- *Job Breakdown: Manual* workers represent 16.7% (1.2K people, tied with Management), which is the highest manual laborer proportion among all active clusters.
- *Age Profile:* Like the Recent segment, this cluster has a heavy concentration of seniors aged **60+** (**29.3%**, 2.1K buyers), alongside a noticeable group of younger adults aged 30–40 (18.1%).
- *Basket Size:* The AOV is very small at **$47.33**, suggesting low-spend purchases. Specific accessory categories should be verified using sales lines and product attributes.

5. **5. Lost / Dormant (Disengaged Lower-Income Manual Workers):**

- *Occupation Disparity: Skilled Manual* workers heavily dominate at **37.79%** (339 buyers), while *Management* drops to its lowest proportion at **8.70%**.
- *Compressed Earnings:* Over **80.49%** of customers earn between $25,000 and $75,000, with only 4.24% earning above $100,000.
- *Complete Inactivity:* With $0.00 in spend and over 600 days without an order, these customers are dormant in the measurement period; permanent churn cannot be proven from a fixed historical snapshot.

## 6.4 Page 3: Customer RFM Segmentation & Behavioral Matrix

Page 3 explains the 5 customer clusters from our K-Means model using the RFM framework (Recency, Frequency, Monetary).

### 6.4.1 Company-Wide Benchmark Metrics

The top ribbon gives the averages for the 30 June 2014 reporting snapshot:

- **Average Recency: 189.32 Days** (around 6.3 months since the last order).
- **Average Frequency: 1.17 Orders** per customer.
- **Average Monetary: $844.76** total spend per customer.

### 6.4.2 Cluster Performance Table

Table 6.2 reproduces the five segment statistics reported by the current dashboard export (subject to final refresh and reconciliation):

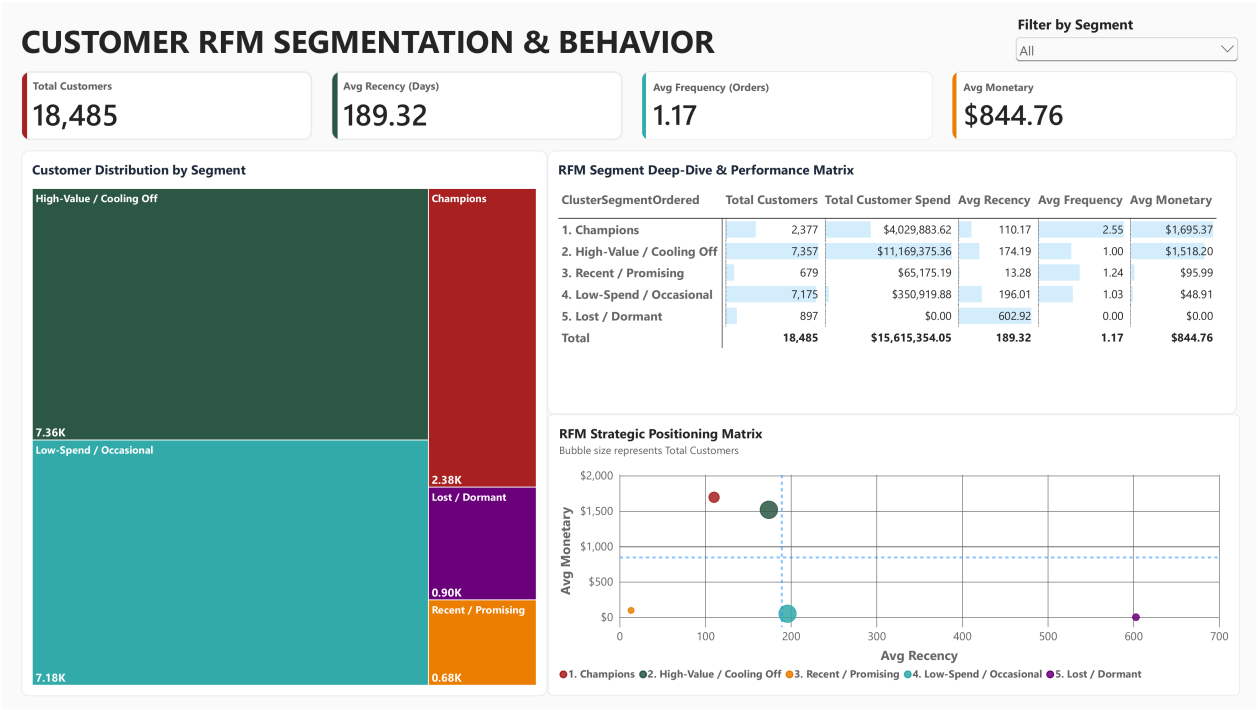



Figure 6.3: Customer RFM Segmentation & Behavior Dashboard (Page 3).

Table 6.2: Customer RFM Cluster Statistics.

| **Cluster Segment** | **Customers** | **Total Spend ($)** | **Avg R (Days)** | **Avg F** | **Avg M ($)** |
|---|---|---|---|---|---|
| 1. Champions | 2,377 (12.86%) | $4,029,883.62 | 110.17 | 2.55 | $1,695.37 |
| 2. High-Value / Cooling Off | 7,357 (39.80%) | $11,169,375.36 | 174.19 | 1.00 | $1,518.20 |
| 3. Recent / Promising | 679 (3.67%) | $65,175.19 | 13.28 | 1.24 | $95.99 |
| 4. Low-Spend / Occasional | 7,175 (38.82%) | $350,919.88 | 196.01 | 1.03 | $48.91 |
| 5. Lost / Dormant | 897 (4.85%) | $0.00 | 602.92 | 0.00 | $0.00 |
| **Company Total / Average** | **18,485** | **$15,615,354.05** | **189.32** | **1.17** | **$844.76** |

### 6.4.3 Customer Volume Distribution (Donut Chart)

The donut chart on the left side of Page 3 illustrates the relative share of each customer group:

- **High-Value / Cooling Off:** 7,357 customers (**39.80%** of total volume).
- **Low-Spend / Occasional:** 7,175 customers (**38.82%**).
- **Champions:** 2,377 customers (**12.86%**).
- **Lost / Dormant:** 897 customers (**4.85%**).
- **Recent / Promising:** 679 customers (**3.67%**).

This chart emphasizes that more than 78% of the customer base is divided almost evenly between two contrasting groups: high-spending buyers who need urgent retention, and low-spending buyers who make infrequent accessory purchases.

### 6.4.4 Strategic Positioning Matrix (Bubble Chart)

The bubble chart at the bottom-right plots all 5 clusters:

- **X-Axis:** Average Recency (from 0 to 700 days). Lower is better.
- **Y-Axis:** Average Monetary (from \$0 to \$2,000). Higher is better.
- **Bubble Size:** Total number of customers in the cluster.
- **Benchmark Lines:** Dotted reference lines at company averages ($X = 189.32$ days and $Y = \$844.76$) divide the chart into four quadrants.

#### Quadrant Analysis

1. **High spend, below-average recency — Champions:** These customers bought recently ($R = 110.17$ days) and spend the most ($M = \$1,695.37$). They are our best VIP customers.
2. **High spend, moderately recent — High-Value / Cooling Off:** These customers spent a lot ($M = \$1,518.20$), but their Recency is 174.19 days, which is close to the 189-day average. They are starting to cool down and need immediate attention before they leave.
3. **Low spend, very recent — Recent / Promising:** These are new customers who bought recently ($R = 13.28$ days). Their spend is still small (\$95.99), but they have great potential to grow into Champions.
4. **Low spend, long recency — Low-Spend and Lost:** These customers have not bought for a long time and spend very little. The Lost group has not returned for over 600 days.

## 6.5 Summary of Core DAX Measures

Table 6.3 records the main measure *logic* described by the team. Names shown here are conceptual; the reporting model derives its table labelled `FactCustomerRFMLive` from the June 2014 snapshot, rather than from the SQL table of the same name. The final Power BI field expressions should be checked during dashboard handover:

Table 6.3: Dashboard measure definitions (conceptual DAX logic).

| **Measure Name** | **Home Table** | **DAX Formula / Logic** |
|---|---|---|
| Total Customers | FactCustomerRFMLive | `DISTINCTCOUNT(CustomerKey)` |
| Total Customer Spend | FactCustomerRFMLive | `SUM(Monetary)` |
| Inactive Customers (>100d) | FactCustomerRFMLive | `CALCULATE([Total Customers], Recency > 100)` |
| Repeat Customer Rate | FactCustomerRFMLive | `VAR R = CALCULATE([Total Customers], Frequency >= 2) RETURN DIVIDE(R, [Total Customers], 0)` |
| Average Order Value (AOV) | FactCustomerRFMLive | `DIVIDE([Total Customer Spend], SUM(Frequency), 0)` |
| Avg Spend per Customer | FactCustomerRFMLive | `DIVIDE([Total Customer Spend], [Total Customers], 0)` |
| Rolling 365-day Monetary Trend | FactCustomerSnapshot | `SUM(Monetary365d)` |

The business actions supported by these visualizations are discussed in Chapter 7.

# Chapter 7

# Business Insights and Decision-Making

## 7.1 From Customer Segments to Business Actions

The three dashboard pages support different but connected questions. The executive page shows the distribution of customer segments over time, the demographic page helps the team understand whom each segment contains, and the RFM page shows how recently and frequently customers buy and how much they spend. These results can be used to prioritize marketing resources rather than sending the same campaign to the entire customer base.

The strategies below have been retained from the team's dashboard analysis. They are *proposed business actions*, not campaigns executed or experimentally proven by this project. Reported numerical values refer to the dashboard's 30 June 2014 analytical snapshot and should be reconciled against the final refresh before presentation.

## 7.2 Actionable Business Recommendations

Based on the dashboard insights, we recommend the following practical strategies for CompanyX:

### 7.2.1 1. Champions: VIP Loyalty Program

- **Goal:** Keep our best customers happy and encourage them to recommend the brand to others.
- **Action:** Create a VIP club that gives early access to new bicycle models, invitations to special cycling events, and referral discounts when they introduce friends.

### 7.2.2 2. High-Value / Cooling Off: Urgent Retention Campaigns

- **Goal:** Protect a segment representing over 71% of the monetary amount in the final reporting snapshot.
- **Action:** Set up an automated email at day 120 after their purchase offering a free annual bicycle check-up at local partner shops, or a $150 discount voucher if they upgrade components within 30 days.

### 7.2.3 3. Recent / Promising: Cross-Selling and Welcome Series

- **Goal:** Encourage new buyers to make a second purchase quickly.
- **Action:** Send a 30-day email series: Day 3 with bike assembly and safety tips, Day 14 with cleaning advice, and Day 21 offering a 15% discount on essential accessories like helmets, gloves, and lights.

### 7.2.4 4. Low-Spend / Occasional: Low-Cost Automated Marketing

- **Goal:** Increase purchase frequency without spending too much marketing budget.
- **Action:** Avoid expensive phone calls or printed mail. Instead, send automated emails and seasonal clearance announcements for cheap consumables like water bottles and spare tubes.

### 7.2.5 5. Lost / Dormant: Cost Reduction

- **Goal:** Avoid costly broad advertising to a low-engagement segment (whose average snapshot recency exceeds 600 days).
- **Action:** Remove this group from paid Facebook and Google ads. Only send them one or two broad holiday sale emails per year (e.g., Black Friday or New Year).

## 7.3 Business Decision-Making Use Case

Consider a marketing team with a limited campaign budget. The dashboard shows that the *High-Value / Cooling Off* segment contains 7,357 customers and represents a large share of the June 2014 snapshot's monetary value. The team can use the segment slicer to inspect the associated demographics, then prioritize a time-limited maintenance or accessory follow-up campaign for this group. A second, low-cost welcome campaign can target *Recent / Promising*

buyers, whose small initial spend and low recency suggest an opportunity to encourage a further purchase.

This is a *decision-support use case*, not evidence that either campaign was executed or increased revenue. An eventual campaign evaluation would compare actual response, incremental sales, and campaign costs against a suitable baseline. The dashboard therefore demonstrates how the segmentation provides a targetable business action while keeping the distinction between descriptive analysis and realized business impact.

**Interpretation note.** The dashboard's figures reflect the current saved report and reference-date filters. The ML chapter contains an illustrative 18,484-customer model extract, whereas this dashboard reports 18,485 customers. The team should reconcile that one-record discrepancy during the final dashboard refresh before asserting identical populations across the two exports.

# Chapter 8

# System Integration, Verification and Limitations

## 8.1 End-to-End Integration

The solution connects the operational `CompanyX` database to a customer-focused SQL Server warehouse through SSIS. The customer-profile package loads newly observed or changed demographic records into staging and applies Type 0/1 and Type 2 rules. The live RFM package updates affected customers and executes the Python scoring process with the latest active K-Means configuration. A separate monthly package appends historical RFM observations as of the previous month-end and applies model-based segment labels. Annual training produces stored model configurations from completed year-end snapshots. Power BI reads the warehouse for executive, demographic, and behavioral analysis.

This division avoids recomputing the full customer population after every transaction. It also deliberately distinguishes two reference dates: `FactCustomerRFMLive` reflects a customer's most recent purchase event, whereas `FactCustomerSnapshot` evaluates the customer population at a shared historical month-end date. Thus, monthly snapshots support inactivity and trend analysis; a customer who makes no new purchase is not automatically rescored in the live table simply because time passes.

## 8.2 Operating Schedule and Dependencies

The team reports that the deployed customer-profile and live RFM SSIS packages execute every minute. The monthly snapshot package runs at 00:05 on the first day of the month, computing the previous month's closing snapshot. The annual Airflow DAG is written to run on 1 January. Its automatic execution, however, was not tested end to end. The present DAG

schedule precedes the monthly snapshot by five minutes, so reliable annual training requires a dependency or readiness check that confirms the year-end snapshot is committed before training begins. Until that dependency is verified, the automated year-end handoff must be described as configured rather than demonstrated.

## 8.3 Verification and Reproducibility

Each stage has outputs that can be independently inspected in SQL Server or the reporting application. Customer extraction and merge can be checked using the number of current customer profiles and the effective-date history in `DimClusterProfiling`. Month-end snapshot runs can be checked by `SnapshotDateKey`, row count, and the absence of unintended duplicate customer/date combinations. Model training can be checked by the presence of five persisted segment configurations for each completed model version. The Python live scorer's output can be checked through the saved `ClusterSegment` and `ModelConfigKey` of affected customer rows. Finally, the Power BI model can be checked against the 30 June 2014 snapshot used for its reference population.

These checks describe the team's testable acceptance criteria; they should not be mistaken for a published, independently measured performance benchmark. The report does not claim quantified improvements in throughput, churn, revenue, or advertising effectiveness without a corresponding controlled comparison.

## 8.4 Known Limitations

- **Purchase-time live semantics.** A live record is refreshed when its customer makes a purchase. Its recency is zero at that purchase time, so the live table alone is not a continuously updated picture of customer inactivity. Monthly snapshots address a different question, evaluating inactivity at common dates.

- **Historical reconstruction.** The first model was established through the 2011 bootstrap and used retrospectively for earlier snapshots. These labels were not predictions actually available at those historical dates. Historical profile attributes may also reflect the available warehouse history rather than a complete reconstruction of past demographics.

- **Annual orchestration.** The Airflow training DAG is implemented but its end-to-end execution and snapshot-readiness dependency remain unverified.

- **Reporting reconciliation.** The saved ML example and dashboard have a one-customer population difference (18,484 versus 18,485), and Page 1 is still subject to final dashboard

refinements. The team should reconcile the extract/filter definitions before citing the counts as identical.

- **Business outcomes.** The dashboard provides decision support and proposed campaign use cases. The group has not run a controlled marketing campaign to establish an improvement in retention, cross-selling, or revenue.
- **Deployment.** The integrated system is operating locally. Deploying the warehouse online remains a stretch goal rather than a delivered feature.

## 8.5 Connection to the Seminar Requirements

The warehouse and SSIS chapters address the customer-focused dimensional design, slowly changing dimensions, incremental loading, and near-real-time processing. The RFM/K-Means chapter documents the data-mining technique and its interpretable outputs. The Power BI and business-decision chapters demonstrate how those outputs are visualized and translated into a concrete customer-retention use case. This integration chapter makes the operating boundaries and remaining deployment work explicit.

# Chapter 9

# Conclusion and Future Work

This project combines customer-focused data warehousing, incremental SSIS integration, versioned K-Means segmentation, and interactive Power BI reporting. The final dimensional design contains two fact tables with complementary purposes: `FactCustomerRFMLive` provides a purchase-triggered operational customer state, and `FactCustomerSnapshot` provides comparable month-end observations. The system also retains demographic changes using slowly changing dimensions and stores model parameters so that customer segments can be assigned without repeating training for each purchase.

Historical models were developed for the 2011, 2012, and 2013 training periods, and five business-oriented customer segments were used to interpret the June 2014 dashboard. The dashboards can support practical decisions such as prioritizing a retention offer for *High-Value / Cooling Off* buyers, or designing lower-cost follow-up messages for *Recent / Promising* customers. These recommendations are analytical use cases; the project did not measure the causal effect of a deployed marketing campaign.

The SSIS packages have been deployed for minute-level customer and live RFM refreshes and monthly historical snapshots. The annual Airflow DAG has been configured, but its unattended execution and dependency on the completed December snapshot have not been tested. Further improvement would include completing the Airflow-to-SSIS dependency, reconciling the model and dashboard customer counts, finalizing the Power BI report, validating the source-derived business statistics, and deploying the warehouse online if time and resources allow. These improvements do not change the customer-centred scope of the completed local implementation.

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